Case Study: Transaction & Customer Behavior Analysis

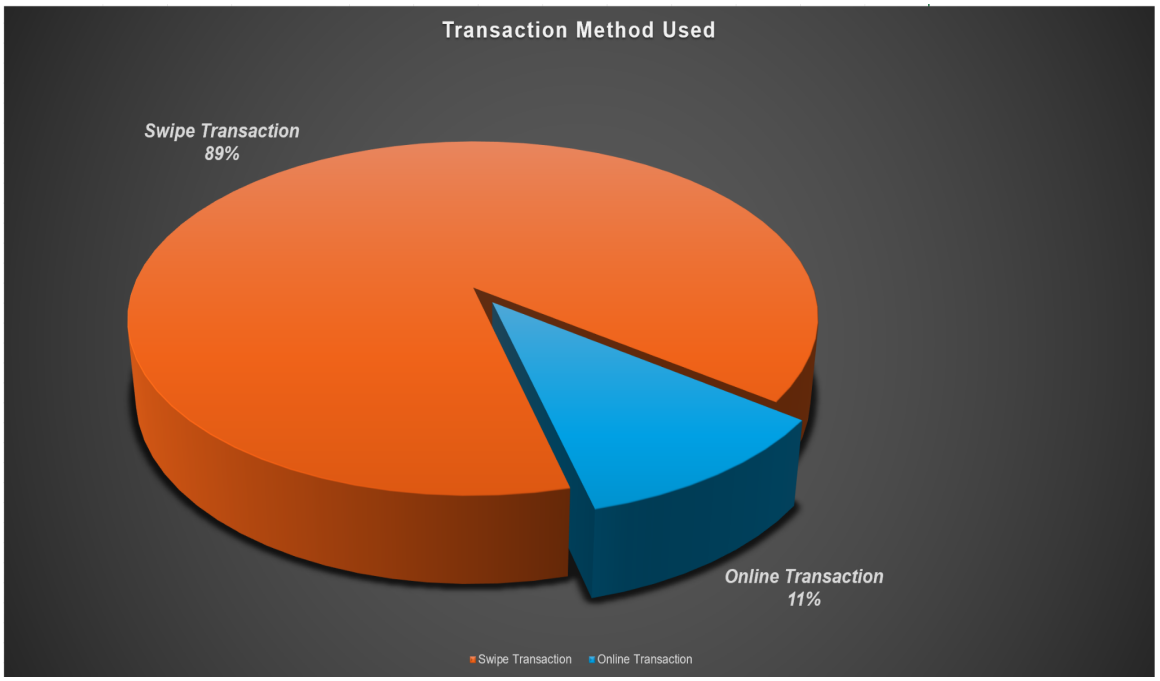
Role Focus: Product Operations | Data Analyst

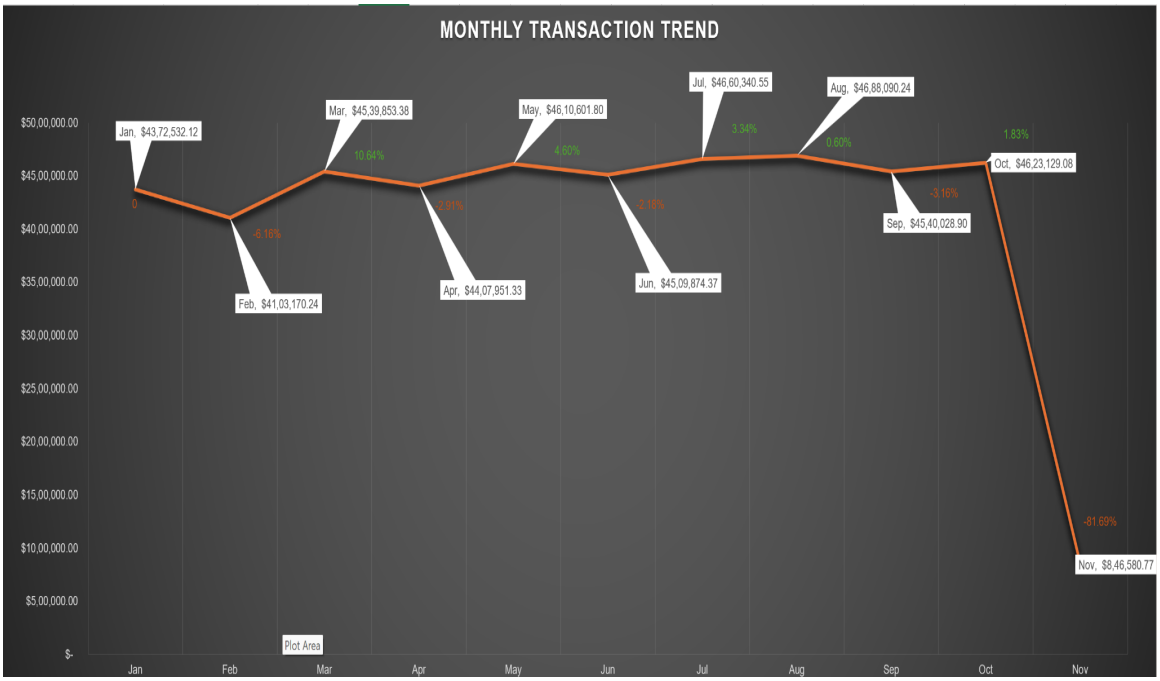
1. Problem Statement

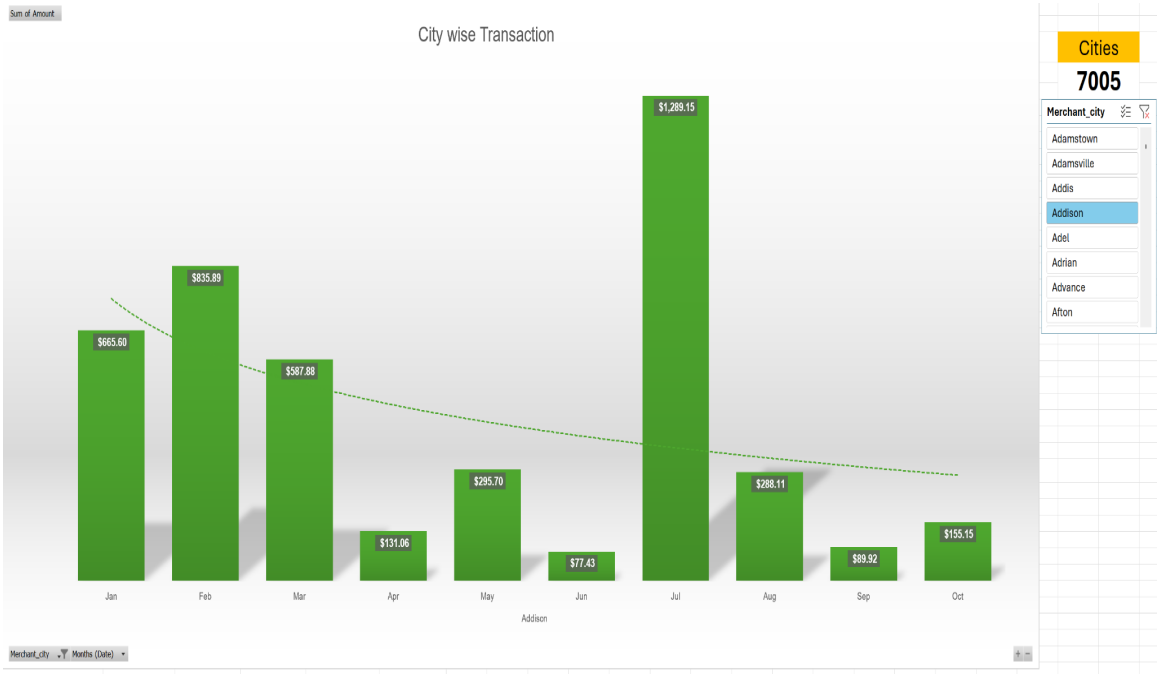
Analyze 1M+ transaction records across 7,005 cities to identify growth opportunities, improve digital adoption, and enhance customer retention.

2. Approach (ETL + Analytics)

Built an end-to-end ETL pipeline using Excel & Power Query. Cleaned, transformed, and structured data. Created automated dashboards for trend, payment, and city-level analysis.







3. Key Insights

• 89% transactions via swipe → low digital adoption • Peak revenue in Aug (~$4.68M) • November drop (-81.69%) → anomaly • High variability across cities • CLV shows revenue concentration in top users

4. Business Impact

Identified key levers for growth: digital payments, retention strategies, and geo-targeted campaigns. Reduced reporting effort via automation.

5. Recommendations

• Incentivize digital payments • Focus campaigns in peak months • Investigate anomalies • Apply city-level targeting • Build retention programs

6. Skills Demonstrated

Advanced Excel | ETL | Data Cleaning | Dashboarding | Business Analysis | Product Thinking